Diffie–Hellman Key Exchange Protocol

28. Write a C program for Diffie-Hellman protocol, each participant selects a secret number x and sends the other participant ax mod q for some public number a. What would happen if the participants sent each other xa for some public number a instead? Give at least one method Alice and Bob could use to agree on a key. Can Eve break your system without finding the secret numbers? Can Eve find the secret numbers?

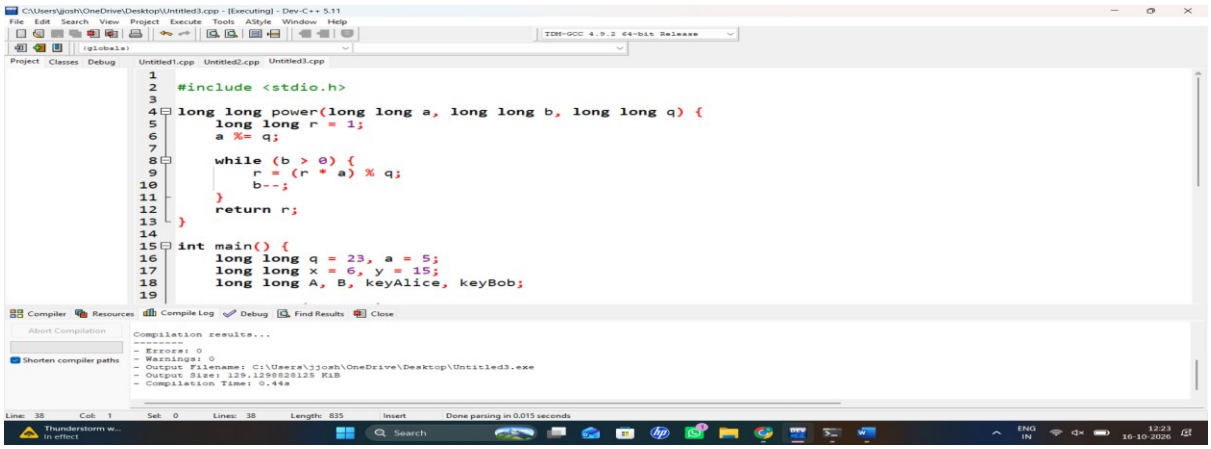
Aim:

To implement the Diffie–Hellman key exchange protocol in C and demonstrate how two participants establish a shared secret key.

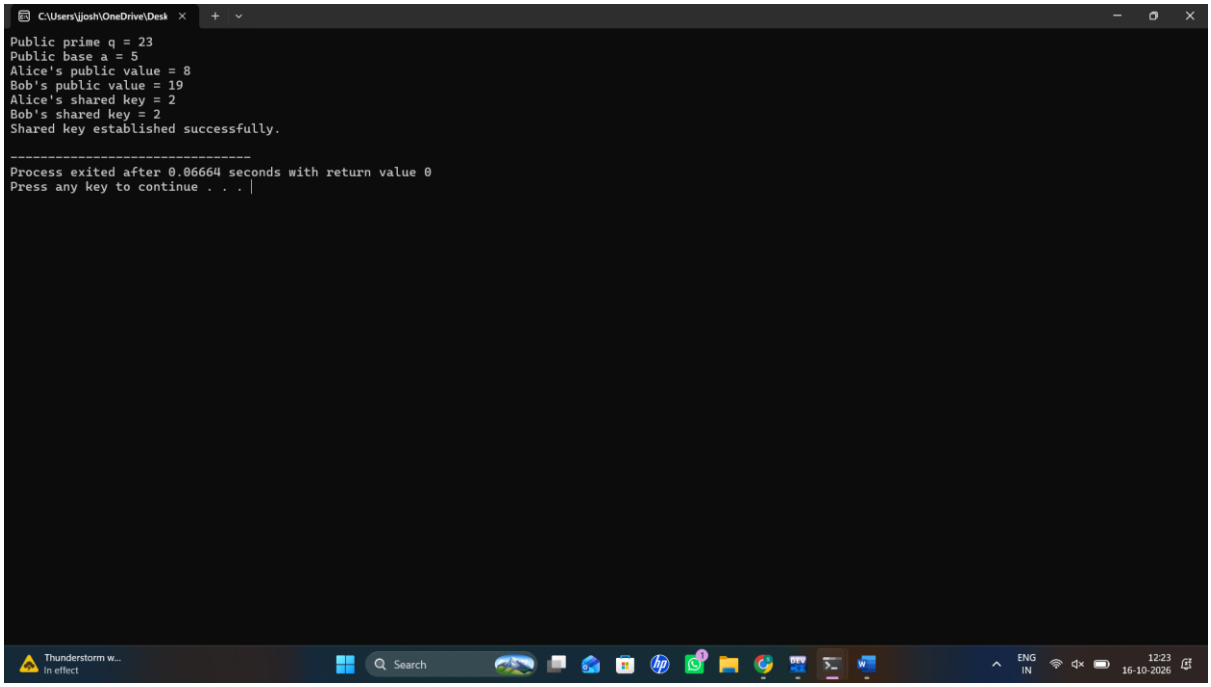
Algorithm:

1. Choose public prime number \(q\) and public base \(a\).
2. Alice selects secret \(x\), and Bob selects secret \(y\).
3. Alice sends \(A=a^x\bmod q\) to Bob.
4. Bob sends \(B=a^y\bmod q\) to Alice.
5. Alice calculates \(K=B^x\bmod q\).
6. Bob calculates \(K=A^y\bmod q\).
7. Verify that both participants obtain the same shared key.

Program:



Output:



Result:

The Diffie–Hellman protocol was implemented successfully, and both participants obtained the same shared secret key.

Security explanation:

If participants send \(x^a\) and \(y^a\) instead of \(a^x\bmod q\), they can still agree on \(K=x^ay^a\bmod q\), but the exchanged values do not provide the usual security of Diffie–Hellman. For example, if \(a=2\), Eve can compute the shared key by multiplying the exchanged values modulo \(q\). In the usual protocol, Eve generally faces the discrete logarithm and computational Diffie–Hellman problems. Authentication is also required to prevent man-in-the-middle attacks.